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Minh-Hai Nguyen

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Summary

I am a 3rd-year PhD candidate in MAMBO team at Toulouse University (expected graduation: Dec. 2026), advised by Prof. Pierre Weiss and Prof. Edouard Pauwels. My research focuses on computational imaging with deep learning and generative models. I also actively contribute to open-source software.

Work Experience

Research Engineer – Agenium Space Dec 2022 - Sept 2023

Correcting micro-vibrations in satellite imaging with pushbroom cameras. Modeling the imaging process, implementing deep learning-based methods, and evaluating them on real data from Sentinel-2B.

Research Intern – Toulouse Mathematics Institute Jun 2022 - Sept 2022

Product-convolution neural networks: applications to image deblurring with space-varying blurs, with Pierre Weiss and Paul Escande.

Education

2023 - present	PhD (Applied Mathematics, Inverse problems in Imaging) at Toulouse University, France	
2018 - 2023	Master of Engineering in Applied Mathematics at INSA Toulouse, France	(Major in AI / Data Science)
2015 - 2018	Quoc Hoc Hue High School for Gifted Students, Vietnam	(Specialized in Math)

Open-source Projects

Blind deblurring in Fluorescence Microscopy Online demo

Designing deep learning-based methods to jointly estimate the sharp image and the unknown blur kernel from a single blurred fluorescence microscopy image. We address diffraction-limited blurs based on Zernike polynomials, spatially varying blurs and 2D/3D settings. Efficient implementation in PyTorch and ongoing validation on various microscopy settings (RIM, TIRF, confocal). Joint work with Pierre Weiss, Florian Sarron, Paul Escande and opticians Thomas Mangeat, Sylvain Cantaloube. Presented at MIFOBIO-25.

DeepInverse – Core contributor and maintainer Link to Github

An open-source PyTorch-based library (part of the PyTorch Ecosystem) for solving imaging inverse problems with state-of-the-art deep learning based methods. `deepinv` accelerates research across imaging domains, enhances research reproducibility via a common modular framework of problems and algorithms. Responsibilities: core library API design/contribution, benchmarks, CI/CD, packaging, docs, and community support.

Skills

Programming	Python: PyTorch (advanced), GPU programming, triton/cuda kernel (intermediate), JAX. C++ (intermediate). Large-scale training (DDP, FSDP) on GPU clusters (Jean-Zay). Profiling/optimization (Nsight, torch-profiler)
Development	Linux, Git/Github, packaging (PyPI), virtual environments (`conda`, `uv`, `docker`). SOTA implementation and evaluation, model training and fine-tuning.
Research	Image restoration, computational imaging, deep learning, deblurring/denoising, generative models, CNNs, Transformers, diffusion/flow models.
Language	English (professional), French (fluent), Vietnamese (native)

Talks and Presentations

- PyTorch Conference Europe 2026 (Speaker) — Paris, France, Apr. 2026
  Title: DeepInverse and Computational Imaging
- Blind inverse problems in imaging: from foundations to applications — Marseille, France, Oct 2025
  Title: How diffusion prior landscapes shape the posterior in blind deconvolution
- GRETSI'25 (Oral) — Strasbourg, France, Aug. 2025
  Title: How diffusion prior landscapes shape the posterior in blind deconvolution
- IABM (French Conference on AI in Medical Imaging) (Poster) — Nice, France, Mar. 2025
  Title: Deep learning-based blind deblurring in microscopy
- 30 years of Mathematical for Optical Imaging (Oral) — Marseille, France, Oct. 2023
  Title: DeepVibes: Correcting Microvibrations in Satellite Imaging With Pushbroom Cameras

Teaching

- Teaching Assistant (Vacataire) – **INSA Toulouse** — 2023 - present
  Courses: Signal Processing, Stochastic Optimization (32 hours per year)

Distinctions

- French National Prize for Open Science Library for DeepInverse — Nov 2024
- Doctoral Research Fellowship, French Ministry of Higher Education and Research — 2023 - 2026
- IA Pau Winner — 2023

Publications

Nguyen, Minh Hai et al. (2026). *A Morse-Bott Framework for Blind Inverse Problems: Local Recovery Guarantees and the Failure of the MAP*. arXiv: 2508.02923 [cs.CV]. URL: https://arxiv.org/abs/2508.02923.

Nguyen, Minh Hai et al. (2026). "An analytic theory of convolutional neural network inverse problems solvers". In: *Proceedings of the 43rd International Conference on Machine Learning (ICML)*. arXiv: 2601.10334 [cs.CV]. URL: https://arxiv.org/abs/2601.10334.

Tachella, Julián et al. (2025). "DeepInverse: A Python package for solving imaging inverse problems with deep learning". In: *Journal of Open Source Software* 10.115, p. 8923. DOI: 10.21105/joss.08923. URL: https://doi.org/10.21105/joss.08923.

Nguyen, Minh Hai and Pierre Weiss (2024). "Comparing Plug-and-Play and Unrolled networks". In: *Pre-print*.

Nguyen, Minh Hai et al. (2024). "DeepVibes: Correcting Microvibrations in Satellite Imaging With Pushbroom Cameras". In: *IEEE Transactions on Geoscience and Remote Sensing* 62, pp. 1–9.